

VENKATA SURYA CHAITANYA PALEPU

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WORK EXPERIENCE

Auditoria.AI

Lead AI Software Engineer, San Jose

May 2024 – Present

- **Led 0→1 development of an AI agent** that serves as an on-demand financial analyst for enterprise AP teams—users ask natural-language questions and receive charts, insights, and auditable answers grounded in their own financial data; shipped to GA as a core feature of Auditoria’s AP automation platform.
- Built **MCP-based agentic tools** for context management, semantic financial modeling, SQL generation, retrieval grounding, and structured outputs.
- Integrated **E2B sandboxes** to safely run isolated financial computations inside the AI agent.
- Built SmartMeta, a **dynamic prompt-management app** with versioning, rapid testing, and seamless deployment for AI workflows (intent detection, document classification, invoice extraction).
- Self-hosted Azure Form Recognizer in an EKS cluster using Azure Cognitive Services containers, ensuring secure and compliant processing of over 1M documents per month at 99.9% uptime.

TCS Research & Innovation Labs

ML Systems Engineer

November 2020 – July 2022

- Designed **high-performance, cost-efficient, and scalable cloud architectures** for inference and training of Machine and Deep Learning (DL) workloads.
- **Published 6 research papers** at top-tier conferences like High Performance and Distributed Computing (HDPC22), and ICPE and filed **2 patents** in the domain of serverless computing over the past 2 years.[Link](#)
- Achieved a significant 30% reduction in execution costs by implementing a **scalable FaaS-based architecture** for document processing DL workflow, leveraging AWS Lambda, Elastic File System, and DynamoDB.[Link](#)
- Devised a **Service Level Agreement (SLA) aware workload scheduling** cloud architecture using AWS Sagemaker and Lambda. This architecture reduced the SLA violation to ~10% for the Deep Learning inference. [Link](#)
- Developed a novel **distributed model training** approach addressing serverless platform limitations, alongside an analytical model to evaluate performance across data management services (S3, Memcached, DynamoDB), resulting in a **1.2x to 5.5x enhancement** in training efficiency and cost-effectiveness. [Link](#)

Oracle Cerner Corporation

Software Engineer

January 2020 – November 2020

- Showcased an impactful Proof-of-Concept **full-stack web application** that streamlined client service onboarding, resulting in a significant reduction of 25 man hours per week.
- **Automated Elastic Map Reduce (EMR) data-processing** pipelines using AWS Step Functions to reduce 70% of manual execution steps. Optimized Spark queries to efficiently **filter millions of health records**.

TECHNICAL SKILLS / CERTIFICATIONS

AI Dev Tools: Cursor, claude code, codex

AI / ML & LLMs: PyTorch, TensorFlow, Hugging Face, LangChain, vLLM, LiteLLM, LLM fine-tuning & deployment

Cloud & MLOps: AWS(SageMaker, Lambda, EKS), Azure, Kubernetes, Docker, Helm, Jenkins, GitHub Actions, CloudFormation

Languages & Frameworks: Python, Java, C + + , SQL, FastAPI, Flask, gRPC, Spring Boot, React

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Master of Science in Computer Science

August 2022 – May 2024

GPA: 3.9/4.0

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science and Engineering

July 2016 – June 2020

GPA: 9.54/10(Ranked 5/250)